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Mamba-FETrack V2: Revisiting state space model for frame-event based visual object tracking

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ABSTRACT

Combining traditional RGB cameras with bio-inspired event cameras for robust object tracking has garnered increasing attention in recent years. However, most existing multimodal RGB-Event tracking algorithms depend heavily on high-complexity Vision Transformer architectures for feature extraction and fusion across modalities. This not only leads to substantial computational overhead but also limits the effectiveness of cross-modal interactions. In this paper, we propose an effective RGB-Event object tracking framework based on the linear-complexity Vision Mamba network, termed Mamba-FETrack V2. Specifically, we first design a lightweight Prompt Generator that utilizes embedded features from each modality, together with a shared prompt pool, to dynamically generate modality-specific learnable prompt vectors. These prompts, along with the modality-specific embedded features, are then fed into a Vision Mamba-based FEMamba backbone, which facilitates prompt-guided feature extraction, cross-modal interaction, and fusion in a unified manner. Finally, the fused representations are passed to the tracking head for accurate target localization. Extensive experimental evaluations on multiple RGB-Event tracking benchmarks, including short-term COESOT dataset and long-term datasets, i.e., FE108 and FELT, demonstrate the superior performance of the proposed tracking framework. The source code and pre-trained models will be released on https://github.com/Event-AHU/Mamba_FETrack.

1. Introduction

Visual Object Tracking (VOT) is a crucial task in the field of computer vision, aiming to locate a given target in subsequent video frames, given its initial position in the first frame. This task demonstrates significant practical value, covering a wide range of important fields such as security surveillance, autonomous driving perception, sports analytics, and human-computer interaction. Currently, most visual object tracking algorithms (He et al., 2024; Tang & Ling, 2019; Wang et al., 2015) are designed and developed based on RGB cameras. With the continuous advancement of deep learning technologies, these unimodal tracking approaches have demonstrated exceptional performance in specific scenarios. Despite the good performance achieved, RGB cameras still have certain limitations. In conditions such as low light, overexposure, and fast motion, the data captured by RGB cameras often performs poorly, affecting tracking accuracy.

To overcome the inherent limitations of conventional RGB cameras, researchers have begun to combine other auxiliary modalities to address challenges under extreme conditions, leading to the emergence of bio-inspired event cameras (e.g., DVS 346, Prophesee, CeleX-V). Unlike the RGB cameras, event cameras perceive information by capturing changes in the light intensity of a scene. When the light intensity changes exceed a given threshold, the event camera generates an event signal at the corresponding pixel location. This event signal is represented by a quadruple $\{x, y, t, p\}$, where (x, y) denotes the spatial location of the event, t represents the timestamp of the event, and p indicates the polarity of the light intensity change. Due to the unique data acquisition principle of event cameras, they offer advantages such as high dynamic range, high temporal resolution, and low power consumption. Consequently, event cameras often demonstrate significant advantages over frame-based RGB cameras, particularly in challenging environments where the latter struggle.

Although event cameras have the advantages mentioned above, they also have certain limitations, such as an inability to capture fine texture details of targets. Fortunately, this is exactly where RGB cameras excel. Therefore, the idea of combining the strengths of both event cameras

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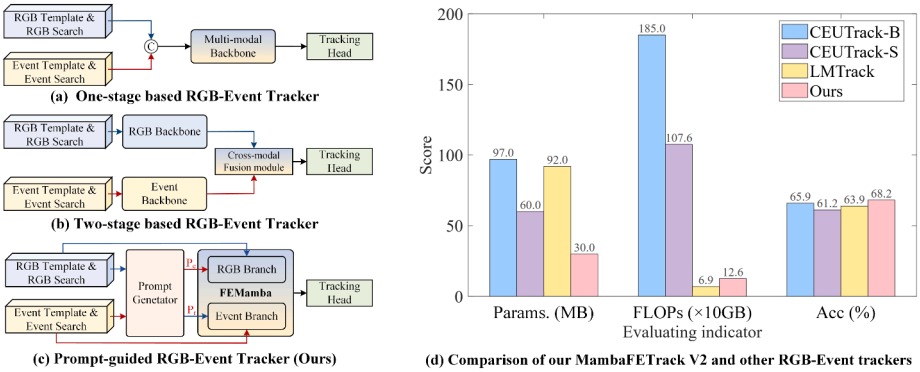


Fig. 1. Architectural comparison of (a) one-stage based and (b) two-stage based trackers, as well as (c) our proposed prompt-guided RGB-Event tracker. (d) A comparison between our proposed Mamba-FETrack V2 method and other robust RGB-Event tracking models in terms of parameters, FLOPs, and accuracy.

and RGB cameras for collaborative visual object tracking has gradually gained increasing attention from researchers, as shown in Fig. 1(a) and (b). For example, Tang et al. (2026) propose a one-stage network architecture for unified tracking, which simultaneously achieves feature extraction, fusion, matching, and interactive learning. Zhang et al. (2023) propose a framework for high frame rate tracking-event alignment and fusion network, which significantly improves the performance of high frame rate tracking by combining the advantages of traditional RGB cameras and event cameras.

Despite the progress achieved by existing RGB-Event tracking methods, they still suffer from two major limitations. *1) high computational complexity remains a critical issue.* Most existing multimodal tracking methods rely on CNNs or ViTs for feature extraction and fusion. CNN-based models are limited in global dependency modeling, while ViT-based models usually introduce high computational complexity due to the self-attention mechanism. *2) modality heterogeneity is still insufficiently addressed.* RGB and event modalities exhibit a deep semantic gap in learned feature spaces. RGB features are usually dense and appearance-dominated, encoding texture, color, and contextual semantics, whereas event features are sparse and change-driven, emphasizing brightness changes, motion boundaries, and high-dynamic-range cues. However, traditional fusion strategies, such as feature concatenation, element-wise addition, and mid-level fusion, usually regard the two modalities as already aligned representations. Directly fusing such heterogeneous features may introduce semantic conflicts, weaken critical modality-specific information, and ultimately degrade model performance.

To mitigate the high computational complexity of existing models, State Space Models (SSMs) (Gu et al., 2022), particularly Visual Mamba architectures (Hatamizadeh & Kautz, 2025; Liu et al., 2024; Zhu et al., 2024), provide a promising solution for RGB-Event tracking. Compared with Transformer-based self-attention, Mamba achieves linear computational complexity while preserving long-range dependency modeling capability, making it well suited for efficiently processing long token sequences. Meanwhile, prompt learning offers an effective mechanism for handling modality heterogeneity. Instead of directly forcing heterogeneous RGB and event features into a shared feature space, prompt mechanisms can encode modality-specific information into learnable guidance vectors, which act as adaptive intermediaries between modalities. Such modality-aware prompts can facilitate selective and context-dependent cross-modal interaction, thereby helping alleviate semantic conflicts and improve the effectiveness of RGB-Event fusion.

In this work, we propose Mamba-FETrack V2, a Vision Mamba-based RGB-Event visual object tracking method. The primary objective of our framework is to achieve effective RGB-Event complementary representation learning while reducing the computational complexity introduced by conventional ViT-based multimodal trackers. Specifically, as shown in Fig. 1(c), inspired by prompt learning (Liu et al., 2023), we design a novel Prompt Generator that encodes the information from RGB and event modalities to generate two modality-specific, learnable prompts, effectively capturing the unique characteristics of each modality. Subsequently, we introduce a unified and more compact Vision Mamba-based backbone network designed to simultaneously perform cross-modal feature extraction, fusion, and interaction. To enable effective interaction between modalities, inspired by MambaIRv2 (Guo et al., 2025), we introduce an innovative state space integration strategy, in which the prompt of each modality is additively fused with the state matrix of the counterpart modality within the SSM module. This design facilitates effective cross-modal information propagation through learned prompt representations. Compared with existing ViT-based methods, it better exploits complementary RGB-Event information while reducing computational complexity and enhancing cross-modal interaction. As shown in Fig. 1(d), our proposed method achieves a good balance between model complexity and accuracy.

To sum up, the main contributions of this paper can be summarized into the following three aspects:

1) We propose Mamba-FETrack V2, a unified Vision Mamba-based RGB-Event tracking framework that integrates feature extraction, cross-modal interaction, and modality fusion within a compact state-space backbone.
2) We introduce a novel input-adaptive Prompt Generator that dynamically produces modality-specific RGB/event prompts from a shared prompt pool, and further propose a prompt-guided state-space fusion strategy that injects these prompts into the counterpart SSM branch to enable efficient cross-modal interaction.
3) We demonstrate the effectiveness of our proposed method through extensive experiments on datasets such as FE108, FELT, and COESOT.

This paper is an extension of our Mamba-FETrack (Huang et al., 2024), which was published on Pattern Recognition and Computer Vision (PRCV) 2024. Although Mamba-FETrack demonstrated the feasibility of introducing Mamba into RGB-Event tracking, its two-stage design still decoupled unimodal feature extraction from cross-modal fusion, leading to architectural redundancy and compromising the compactness of the framework. In addition, its fusion strategy largely depends on rigid feature alignment and direct cross-modal interaction, which may struggle to bridge the semantic gap between dense RGB features and sparse event features, thus limiting flexible cross-modal complementary representation learning. Motivated by these limitations, this paper makes the following major extensions. **(I). New feature modeling framework:** We replace the two-stage feature modeling pipeline with a unified FEMamba backbone, which integrates feature extraction, cross-modal interaction, and fusion within a single compact architecture. **(II). More effective feature fusion and interaction strategy:** We employ a Prompt Generator to produce modality-specific prompts and introduce a prompt-guided cross-fusion mechanism, which replaces the previous FusionMamba module and enables more effective cross-modal feature

interaction. **(III). More comprehensive experiments:** We conduct more detailed and comprehensive experiments to demonstrate the effectiveness of the proposed method. Building upon the original framework, our Mamba-FETrack V2 achieves significant improvements in both efficiency and robustness for object tracking.

The remainder of this paper is organized as follows: The related works are provided in Section 2, focusing on RGB-Event based Tracking, Mamba Network, and Prompt-based Learning. For the proposed approach, we first give an overview in Section 3.1, then dive into the details of input representation, Prompt Generator module, FEMamba backbone network, tracking head, and loss function in subsequent sub-sections. In Section 4, we conduct the experiments on multiple benchmark datasets and both the qualitative and quantitative experiments demonstrate the effectiveness of our proposed Mamba-FETrack V2 framework for RGB-Event based visual object tracking. Finally, we conclude this paper and propose possible research directions for the Mamba-based visual object tracking in Section 5.

2. Related works

In this section, we mainly introduce the related works on the RGB-Event-based Tracking, Mamba Network, and Prompt-based Learning. More related works can be found in the surveys (Lei et al., 2024; Wang et al., 2024b; Zhang et al., 2024a,b) and the paper list.[1]

2.1. RGB-event based tracking

In recent years, event cameras have been receiving increasing attention, and multimodal object tracking algorithms (Hong et al., 2024; Hou et al., 2024; Hu et al., 2025; Liu et al., 2025) that integrate RGB cameras and event cameras have demonstrated strong performance. Tang et al. (2026) propose CEUTrack, which models the relationship between RGB frames and event voxels using a unified Transformer network. Zhang et al. (2021) propose a multimodal tracking approach that combines frame and event domains and balances the contributions of both domains. Wang et al. (2021b) propose a reliable object tracking method through the collaboration of two different sensors (RGB camera and event camera). TENet (Shao et al., 2025) designs a targetness entanglement strategy to promote the selective enhancement of object-related cues in both modalities. Zhu et al. (2023b) propose a mask modeling strategy that randomly masks tokens across modalities to encourage the Vision Transformer (ViT) to close the distribution gap between them, thus improving the model's capability. Despite these advances, the high computational complexity of Transformer networks imposes a heavy computational burden on tracking algorithms. Unlike the aforementioned methods, we are inspired by the linear complexity advantage of the Mamba model and propose a Vision Mamba-based RGB-Event object tracking algorithm.

2.2. Mamba network

Recently, State Space Models (SSMs), initially designed for natural language processing tasks, have gained increasing attention. Related studies have also explored memory-aware mathematical modeling of complex dynamical processes using fractional-order formulations in various physical systems (Abbas et al., 2024a,b, 2026a,b). Mamba (Gu & Dao, 2024) introduced a time-varying SSM equipped with a selective mechanism, demonstrating strong capability in modeling long sequences. Its success has inspired researchers to explore its application across a broad range of fields. In the field of computer vision, Vision Mamba (Zhu et al., 2024) applies a bidirectional SSM design to the field of vision, achieving competitive results in tasks such as image classification, object detection, and semantic segmentation. VMamba (Liu et al., 2024) introduces the Cross-Scan Module (CSM) to traverse spatial dimensions and reorganize non-causal visual inputs into ordered patch sequences.

Hatamizadeh and Kautz (2025) propose MambaVision, a hybrid visual backbone network that combines Mamba and Transformer structures, enhancing its ability to efficiently model visual features. For multimodal learning, Coupled Mamba (Li et al., 2024) proposes a coupled SSM that enhances both inter- and intra-modal correlation modeling, showing strong fusion capabilities. Xie et al. (2024) propose a novel dynamic feature enhancement framework called FusionMamba for multimodal image fusion tasks. In this work, we leverage the linear computational complexity of the Mamba network to achieve a favorable balance between tracking accuracy and model complexity.

2.3. Prompt-based learning

Prompt learning was initially designed for adapting large-scale pretrained language models to downstream tasks. In the field of computer vision, prompt learning has gradually become an effective approach to improving model performance and reducing computational overhead. For instance, Jia et al. (2022) propose VPT, which introduces a small number of trainable parameters into the input space, and the backbone of the model is kept frozen, achieving efficient model fine-tuning. Zhou et al. (2022) propose CoOp, which aims to introduce a learnable prompt approach to better adapt the powerful and generalized priors of visual language models (such as CLIP Radford et al., 2021) to downstream tasks.

Recent work has begun to incorporate prompting mechanisms into the visual object tracking task to improve tracking accuracy. ViPT, proposed by Zhu et al. (2023a), which learns modality-related cues using prompt learning for visual object tracking. Shi et al. (2024) generate explicit visual cues to facilitate inference in the current frame. Different from the above works, in this paper, we design a novel Prompt Generator to generate two modality-specific learnable prompts from both RGB and event modalities, enabling a simple yet effective multimodal fusion approach for visual object tracking.

3. Methodology

3.1. Overview

In this section, we will give an overview of the proposed Mamba-FETrack V2 framework, designed for visual object tracking by effectively fusing information from RGB frames and event streams. As illustrated in Fig. 2, the architecture begins by processing the input RGB template/search region and the corresponding event stream representations. These inputs are transformed into token sequences and added with position encodings. A key component of the framework is the novel Prompt Generator, which dynamically creates modality-specific prompts, denoted as P_r for RGB and P_e for event modality, based on the initial visual representations. Subsequently, a compact FEMamba network is employed to perform joint feature extraction, fusion, and interaction. The generated prompts P_r and P_e are integrated directly into the core computation of two parallel Vision Mamba blocks, effectively steering the feature extraction process and facilitating cross-modal interaction. The resulting enhanced features are concatenated and subsequently fed into the tracking head to predict the target object's location using the response maps. In addition, we also employ the dynamic template mechanism to handle the RGB-Event long-term visual tracking task.

3.2. Input representation

The Mamba-FETrack V2 framework takes as input two primary modalities: a sequence of RGB video frames, denoted as $I = \{I_1, I_2, \ldots, I_N\}$, and the corresponding asynchronous event streams from

[1] github.com/wangxiao5791509/Single_Object_Tracking_Paper_List

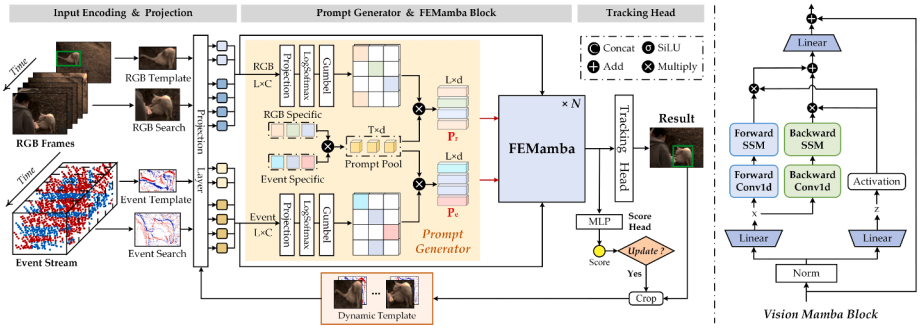


Fig. 2. An overview of our proposed Mamba-FETrack V2 framework. Given the RGB frames and Event streams, we first extract the template and search regions of dual modalities and transform them into tokens. Then, the newly proposed Prompt Generator is adopted to produce the modality-specific prompts dynamically. The FEMamba network is proposed to achieve feature extraction, fusion, and interaction for enhanced RGB-Event feature learning. These features are concatenated and fed into the tracking head for localization of the target object. We also introduce the dynamic template update scheme to handle the issue of appearance variation in the tracking task.

the event camera, represented as $E = \{e_1, e_2, \ldots, e_M\}$. Following standard practice in visual tracking (e.g., OSTrack Ye et al., 2022, Mamba-FETrack Huang et al., 2024), we extract the target template $Z_I \in \mathbb{R}^{B\times3\times128\times128}$ and the search region $X_I \in \mathbb{R}^{B\times3\times256\times256}$ from the RGB frames. To enable processing analogous to that of RGB frames, the sparse and asynchronous event streams are converted into a dense representation suitable for standard visual models. Specifically, we stack the event stream occurring within the exposure time corresponding to each RGB frame, generating an aligned event frame. From these event frames, we similarly extract the event template $Z_E \in \mathbb{R}^{B\times3\times128\times128}$ and event search region $X_E \in \mathbb{R}^{B\times3\times256\times256}$, respectively.

These four images (Z_I, X_I, Z_E, X_E) are first divided into a sequence of non-overlapping small patches. Then, a projection layer (Patch Embedding) maps each small patch to a fixed-dimensional token representation. To preserve crucial spatial information, learnable absolute position encodings are added to the token sequences corresponding to the template and search regions, respectively. This process yields four initial token sequences, namely RGB template tokens $H_I^z \in \mathbb{R}^{B\times N_z\times C}$, RGB search tokens $H_I^x \in \mathbb{R}^{B\times N_x\times C}$, event template tokens $H_E^z \in \mathbb{R}^{B\times N_z\times C}$, and event search tokens $H_E^x \in \mathbb{R}^{B\times N_x\times C}$, where B is the batch size, C is the embedding dimension, and N_z, N_x are the number of tokens for template and search regions, respectively. These sequences serve as the input to the subsequent Prompt Generator and FEMamba backbone networks.

3.3. Prompt generator

To facilitate adaptive guidance and interaction between multimodal features, we introduce the Prompt Generator module. Its core function is to dynamically generate modality-specific learnable prompt vectors based on the input RGB and event embeddings. This module consists of three key components: learnable prompt pools, routing-based prompt selection, and final prompt generation, as shown in Fig. 2.

First, we define two learnable embedding matrices as the basis for modality-specific prompts: $W_r \in \mathbb{R}^{T\times d}$ and $W_e \in \mathbb{R}^{T\times d}$, where T is the number of basis prompts and d is the dimension of each basis prompt. These matrices are combined through element-wise multiplication to construct a shared, learnable prompt pool $P_{pool} \in \mathbb{R}^{T\times d}$:

$$P_{pool} = W_r \odot W_e. \tag{1}$$

This approach potentially introduces beneficial non-linear interactions beyond simpler operations such as concatenation or addition, offering a richer semantic basis for prompt selection.

Next, a routing-based prompt selection mechanism allows the model to dynamically choose relevant prompts from the prompt pool based on the input modality feature. Concretely, the concatenated RGB tokens $H_{rgb} = [H_I^z, H_I^x] \in \mathbb{R}^{B\times L\times C}$ and event tokens $H_{event} = [H_E^z, H_E^x] \in \mathbb{R}^{B\times L\times C}$ (where $L = N_z + N_x$) are passed through a shared routing network F_{route}. This routing network consists of an MLP with LogSoftmax function, and outputs log-probability distributions $R_r^{log} \in \mathbb{R}^{B\times L\times T}$ and $R_e^{log} \in \mathbb{R}^{B\times L\times T}$, which indicate the relevance of each token to each basis prompt in the pool:

$$R_r^{log} = F_{route}(H_{rgb}), \tag{2}$$

$$R_e^{log} = F_{route}(H_{event}). \tag{3}$$

To make the routing process differentiable, we apply the Gumbel-Softmax trick (Tucker et al., 2017) (the temperature is set to a fixed value of 1.0) to the log-probabilities, enabling the generation of one-hot routing matrices $O_r \in \mathbb{R}^{B\times L\times T}$ and $O_e \in \mathbb{R}^{B\times L\times T}$:

$$O_r = GS(R_r^{log}), \tag{4}$$

$$O_e = GS(R_e^{log}). \tag{5}$$

Here, the GS denotes the Gumbel-Softmax operation. Each row of these one-hot matrices effectively identifies the prompt associated with each input token, thereby facilitating adaptive prompt selection based on the current visual representation.

Finally, in the dynamic prompt generation stage, the resulting routing matrices are matrix-multiplied with the shared prompt pool P_{pool} to produce the final dynamic prompt vectors $P_r \in \mathbb{R}^{B\times L\times d}$ and $P_e \in \mathbb{R}^{B\times L\times d}$ for the RGB and event modalities, respectively. The formula can be written as:

$$P_r = O_r \times P_{pool}, \tag{6}$$

$$P_e = O_e \times P_{pool}. \tag{7}$$

These dynamically generated prompts encapsulate informative features that are carefully tailored to the unique characteristics of each input modality. By capturing context-aware and modality-relevant representations, these prompts enhance the model's ability to focus on salient information adaptively. Once generated, these learnable prompts are integrated into the FEMamba backbone, where they facilitate feature fusion and cross-modal interaction between the two modalities. Specifically, the event prompt P_e guides RGB feature extraction, while the RGB prompt P_r symmetrically guides event feature extraction in the FEMamba backbone.

3.4. FEMamba backbone

To effectively extract modality-specific features and facilitate deep cross-modal interaction, we propose a unified FEMamba backbone network, which consists of two parallel Vision Mamba blocks sharing the same architecture, as shown in the right figure of Fig. 3. Compared to the previous version (the left figure of Fig. 3), we employ prompts to enable a simpler yet more effective feature interaction: the RGB backbone learns the RGB features H_{rgb} by incorporating the event prompt P_e,

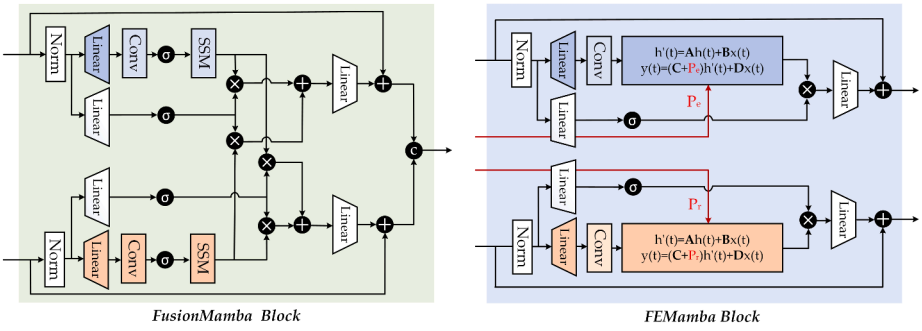


Fig. 3. A comparison between FusionMamba block (Proposed in our conference paper Huang et al., 2024) and the newly proposed FEMamba block.

whereas the event backbone extracts the event features H_{event} guided by the RGB prompt P_r. In the following, we take the RGB branch as an example for clarity.

Let's detail the process within the lth FEMamba block of the RGB branch, which receives $H_{rgb}^{(l-1)}$ and the event prompt P_e as inputs. First, the input tokens are normalized through the Normalization layer, and using parallel linear projections, generate intermediate representations z_{rgb} and x_{rgb}. It can be mathematically represented as:

$$z_{rgb} = \text{Linear}_z(\text{Norm}(H_{rgb}^{(l-1)})), \tag{8}$$

$$x_{rgb} = \text{Linear}_x(\text{Norm}(H_{rgb}^{(l-1)})). \tag{9}$$

The x_{rgb} representation is processed through a 1D depth-wise convolution followed by a SiLU activation to produce x'_{rgb}:

$$x'_{rgb} = \text{SiLU}(\text{Conv1d}(x_{rgb})). \tag{10}$$

Subsequently, x'_{rgb} is linearly projected to generate the state space matrices, B, C, and Δ:

$$B, C, \Delta = \text{Linear}(x'_{rgb}). \tag{11}$$

Using Δ, the discrete state matrix $\bar{A}$ and input matrix $\bar{B}$ are obtained via the Zero-Order Hold (ZOH) discretization rule applied to the base state space matrices A and B:

$$\bar{A}, \bar{B} = \text{ZOH}(A, B, \Delta). \tag{12}$$

Here lies the core prompt-guided cross-modal fusion (PCMF) step. Prior to executing the selective scan, the dynamic event prompt P_e generated based on event features is added to the original output matrix C in the RGB branch. This creates a cross-modality enhanced output matrix, avoiding redundant networks for interactive learning. Therefore, the formula of SSM in the RGB branch can be denoted as:

$$h'(t) = \bar{A}h(t) + \bar{B}x(t), \tag{13}$$

$$y(t) = (C + P_e)h'(t) + Dx(t), \tag{14}$$

where $h(t)$ and $h'(t)$ denote the hidden state at the previous time step and the current moment, respectively. $x(t)$ and $y(t)$ are the input and output at the current moment. Through this mechanism, the event modality features are adaptively transferred to the RGB modality via the event prompt P_e, facilitating natural and effective cross-modal feature interaction learning from event to RGB domains. Integrating the prompt by modifying C is advantageous due to its ease of implementation and validation, negligible computational and memory overhead, and its avoidance of potential gradient issues (e.g., explosion or vanishing) often associated with altering the core SSM state transition dynamics ($\bar{A}, \bar{B}$), thus enabling training stability.

Through selective scanning in both forward and backward directions, we will obtain y_f and y_b, respectively. These outputs are subsequently gated by z and summed:

$$y_{rgb} = y_f \odot \text{SiLU}(z_{rgb}) + y_b \odot \text{SiLU}(z_{rgb}), \tag{15}$$

where SiLU (Sigmoid-weighted Linear Unit) refers to the activation function. Finally, a residual connection yields the block's output. The formula is as follows:

$$H_{rgb}^{l} = H_{rgb}^{(l-1)} + \text{Linear}(y_{rgb}), \tag{16}$$

where the H_{rgb}^{l} represents the output of the RGB branch at the lth layer.

The event branch adopts a symmetric architecture to maintain modality balance. As input, it receives the event token sequence from the previous layer $H_{event}^{(l-1)}$ along with the RGB prompt P_r. Subsequently, a parallel Vision Mamba block implements a prompt fusion mechanism, integrating the RGB prompt P_r into the event modality through $C' = C + P_r$. This operation enables cross-modal guidance from the RGB prompt while preserving the structural integrity of event representations, thereby allowing the distilled RGB semantics to adaptively steer event feature extraction. Through this prompt-guided design, the Prompt Generator and FEMamba backbone collaboratively perform cross-modal feature extraction, interaction, and fusion. The modality-specific prompt generated from one modality is injected into the counterpart SSM branch to guide feature modeling, enabling bidirectional RGB-Event interaction within a unified and compact backbone.

After stacking L such FEMamba blocks with this cross-modal prompt injection, we obtain the final enhanced features $F_{rgb} = H_{rgb}^{L}$ and $F_{event} = H_{event}^{L}$. These features encapsulate rich, modality-specific information that has been mutually enhanced through deeply integrated cross-modal guidance before being forwarded to the tracking head.

To better accommodate the continuous variations in the target's appearance, we empirically integrate a template updating strategy inspired by STARK (Yan et al., 2021). As illustrated in Fig. 2, the framework maintains both an initial static template and an adaptively updated dynamic template. Specifically, the features extracted by the FEMamba backbone are fed into a score prediction head based on an MLP architecture, which estimates the reliability of the current frame. Following STARK, the default update interval and confidence threshold are set to 200 frames and 0.5, respectively. When the update interval is reached and the confidence score exceeds this threshold, the current frame is cropped based on the tracking result, and the dynamic template is subsequently updated. This strategy enables timely adaptation to appearance variations, effectively mitigating performance degradation caused by challenges such as target deformation.

3.5. Tracking head and loss function

Our tracking head design closely follows OSTrack (Ye et al., 2022). The enhanced search region features extracted from the FEMamba backbones, encompassing both RGB and event modalities, are concatenated along the channel dimension. This multimodal feature representation is subsequently processed by the tracking head to estimate the target object's spatial position.

During training, we first adopt a loss structure similar to OSTrack (Ye et al., 2022), employing three distinct loss functions for comprehensive

optimization: Focal Loss ($\mathcal{L}_{focal}$) for classification, L1 Loss ($\mathcal{L}_1$) for offset regression, and GIoU Loss ($\mathcal{L}_{GIoU}$) for bounding box size and overlap regression. The total loss function $\mathcal{L}_{total}$ is a weighted sum of these components:

$$\mathcal{L}_{total} = \lambda_1 \mathcal{L}_{focal} + \lambda_2 \mathcal{L}_1 + \lambda_3 \mathcal{L}_{GIoU}, \tag{17}$$

where $\lambda_1, \lambda_2, \lambda_3$ are weighting coefficients balancing the contribution of each loss term, which are set to 1, 14, and 1, respectively. After training the base model, we follow the approach of STARK (Yan et al., 2021) by incorporating a template updating module to further train the score prediction head. We employ the commonly used Binary Cross-Entropy with Logits Loss (BCEWithLogitsLoss) to supervise the score head. The formulation is given as follows:

$$\mathcal{L}_{\text{bce}} = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \cdot \log \sigma(x_i) + (1 - y_i) \cdot \log(1 - \sigma(x_i))\right], \tag{18}$$

where x_i and y_i denote the predicted logit and the corresponding ground-truth label, respectively, and $\sigma(\cdot)$ represents the sigmoid activation function. These loss functions jointly guide the network to learn accurate target localization and scale estimation.

4. Experiments

4.1. Datasets and evaluation metrics

To comprehensively evaluate the effectiveness of the proposed method, we conducted extensive experiments on three benchmark datasets: FE108 (Zhang et al., 2021), FELT (Wang et al., 2024a), and COESOT (Tang et al., 2026).

• **FE108 Dataset**: This dataset is a dual-modality single object tracking dataset collected using a grayscale DAVIS 346 event camera. It consists of 108 tracking videos captured in indoor scenes, with a total duration of approximately 1.5 h. Among them, 76 videos are used for training and 32 for testing. The dataset covers 21 different object categories, which can be grouped into three main types: animals, vehicles, and daily objects. In addition, it includes four challenging scenario types: low-light (LL), high dynamic range (HDR), fast motion with and without motion blur on APS frame (FWB and FNB). Please refer to the GitHub [2] for more details.

• **FELT Dataset**: This dataset is a large-scale, long-term, dual-modality single-object tracking benchmark, captured using the DAVIS 346 event camera. It contains a total of 1044 video sequences, with 730 sequences used for training and 314 for testing. Each sequence lasts on average more than 1.5 min, ensuring that each video includes at least 1000 synchronized pairs of RGB and event frames. In total, it includes 1,949,680 annotated frames, encompasses 60 object categories, and defines 14 challenging tracking attributes, such as occlusion, fast motion, and low illumination. Please refer to the GitHub[3] for more details.

• **COESOT Dataset**: It is a category-wide RGB-Event-based tracking benchmark designed to evaluate the generalization ability of tracking algorithms across diverse object types. It comprises a total of 1354 video sequences spanning 90 object categories, with 478,721 annotated RGB frames. The dataset is systematically divided into 827 training videos and 527 testing videos. To facilitate fine-grained performance analysis, 17 challenging factors, including fast motion, occlusion, illumination variation, background clutter, and scale change, are explicitly defined. Please refer to the GitHub[4] for more details.

For evaluation metrics, we adopt three widely used metrics in tracking tasks: **Precision (PR)**, **Normalized Precision (NPR)**, and **Success Rate (SR)**. Specifically, Precision (PR) represents the proportion of frames where the error between the predicted center position and the ground truth center position is smaller than a predefined threshold (default is 20 pixels). Normalized Precision (NPR) is calculated by measuring the Euclidean distance between the predicted center and the ground truth center, and then scaling it using the diagonal matrix formed by the width and height of the ground truth bounding box for scale normalization. Success Rate (SR) represents the percentage of frames where the Intersection over Union (IoU) between the predicted bounding box and the ground truth bounding box exceeds a given threshold.

4.2. Implementation details

Our Mamba-FETrack V2 framework is built upon the OSTrack architecture (Ye et al., 2022), with a key modification of replacing the original vision Transformer backbone with the more effective Vision Mamba network (Zhu et al., 2024), which significantly enhances the model's efficiency while maintaining competitive accuracy. We stack 24 layers of the proposed FEMamba block as the backbone network to simultaneously perform feature extraction and modality interaction. The model is trained using the AdamW optimizer (Loshchilov & Hutter, 2018), with a learning rate set to 0.0001, a weight decay of 0.0001, and a batch size of 48. To detect whether the target has drifted outside the search region, we monitor the maximum response score in the current frame. If the maximum scores τ remain below 0.3 for k consecutive frames (with k = 8 on FE108 and FELT, and k = 2 on COESOT), we enlarge the cropping ratio of the search region r by a factor of 1.5. This adaptive adjustment helps the tracker to re-locate fast-moving or abruptly shifting targets that may have left the original search region. All experiments are implemented using the PyTorch framework (Paszke et al., 2019) and conducted on a computing server equipped with an AMD EPYC 7542 32-core CPU and an NVIDIA RTX 4090 GPU. Further implementation details are available in our source code.

4.3. Comparison on public benchmark datasets

• **Results on FE108** Table 1 presents a performance comparison between our proposed method and several state-of-the-art (SOTA) approaches on the FE108 dataset. The results demonstrate that our method ranks first among all compared trackers, achieving PR and SR scores of 94.74 and 62.18, respectively. Compared to Mamba-FETrack (Huang et al., 2024), our proposed Mamba-FETrack V2 shows consistent performance gains in both PR and SR metrics, highlighting the effectiveness of our feature fusion strategy.

• **Results on FELT** As shown in Table 2, we compare the experimental results of our method with other SOTA tracking approaches on the FELT dataset. Specifically, the classic ViT-based tracking algorithm OSTrack achieves scores of 52.3, 65.9, and 63.3 in terms of SR, PR, and NPR, respectively. In comparison, our method surpasses OSTrack, achieving 53.8, 68.2, and 65.2 on the same three evaluation metrics, respectively. Furthermore, compared to other competitive trackers such as AsymTrack, our method demonstrates a superior performance, achieving a +1.5% improvement in precision. These results clearly highlight the effectiveness of our approach on this challenging long-term tracking benchmark.

• **Results on COESOT** As shown in Table 3, we report our tracking results on the large-scale RGB-Event tracking dataset COESOT. Our method surpasses popular tracking algorithms like OSTrack and LMTrack, achieving SR and PR scores of 62.6 and 76.9, respectively. These results highlight the superior tracking performance of our method across various scenarios in the dataset. However, when compared to several robust methods built on high-complexity ViT architectures, such as ViPT, our method shows a disadvantage in terms of SR. This indicates that bounding box alignment can be further improved. Unlike Transformer-based trackers, which explicitly model pairwise token relationships through self-attention, Mamba captures spatial interactions more implicitly through sequential scanning. Although this mechanism enables

[2] https://github.com/Jee-King/ICCV2021_Event_Frame_Tracking?tab=readme-ov-file

[3] https://github.com/Event-AHU/FELT_SOT_Benchmark

[4] https://github.com/Event-AHU/COESOT

Table 1
Experimental results (SR/PR) on FE108 dataset.

Tracker	SiamRPN (Li et al., 2018)	SiamFC++ (Xu et al., 2020)	KYS (Bhat et al., 2020)	CLNet (Dong et al., 2020)	CMT-MDNet (Wang et al., 2021b)	ATOM (Danelljan et al., 2019)
SR/PR	21.8/33.5	23.8/39.1	26.6/41.0	34.4/55.5	35.1/57.8	46.5/71.3
Tracker	DiMP (Bhat et al., 2019)	PrDiMP (Danelljan et al., 2020)	CMT-ATOM (Wang et al., 2021b)	CEUTrack (Tang et al., 2026)	Mamba-FETrack (Huang et al., 2024)	Ours
SR/PR	52.6/79.1	53.0/80.5	54.3/79.4	55.58/84.46	58.71/90.95	62.18/94.74

Table 2
Tracking results on FELT dataset.

Trackers	Source	SR/PR/NPR
01. OSTrack (Ye et al., 2022)	ECCV22	52.3/65.9/63.3
02. GRM (Gao et al., 2023)	CVPR23	52.1/65.6/62.9
03. ROMTrack (Cai et al., 2023)	ICCV23	51.8/65.8/62.7
04. ViPT (Zhu et al., 2023a)	CVPR23	52.8/65.3/63.1
05. SeqTrack (Chen et al., 2023)	CVPR23	52.7/66.9/63.4
06. HIPTrack (Cai et al., 2024)	CVPR24	51.6/65.6/62.2
07. ODTrack (Zheng et al., 2024)	AAAI24	52.2/66.0/63.5
08. SDSTrack (Hou et al., 2024)	CVPR24	53.7/66.4/64.1
09. UnTrack (Wu et al., 2024)	CVPR24	53.6/66.0/63.9
10. LMTrack (Kang et al., 2025)	AAAI25	50.9/63.9/61.8
11. AsymTrack (Zhu et al., 2025)	AAAI25	51.9/66.7/62.0
12. ORTrack (Wu et al., 2025)	CVPR25	48.4/61.7/59.2
13. UNTrack (Qin et al., 2025)	CVPR25	50.0/63.9/61.6
14. Mamba-FETrack (Huang et al., 2024)	PRCV24	52.2/66.7/63.6
15. Ours	–	53.8/68.2/65.2

Table 3
Tracking results on COESOT dataset.

Trackers	Source	SR	PR
01. DiMP50 (Bhat et al., 2019)	ICCV19	58.9	72.0
02. ATOM (Danelljan et al., 2019)	CVPR19	55.0	68.8
03. STARK (Yan et al., 2021)	ICCV21	56.0	67.7
04. TransT (Chen et al., 2021)	CVPR21	60.5	72.4
05. OSTrack (Ye et al., 2022)	ECCV22	59.0	70.7
06. MixFormer (Cui et al., 2022)	CVPR22	55.7	66.3
07. ToMP50 (Mayer et al., 2022)	CVPR22	59.8	70.8
08. ToMP101 (Mayer et al., 2022)	CVPR22	59.9	71.6
09. TrDiMP (Wang et al., 2021a)	CVPR21	60.1	72.2
10. MDNet (Wang et al., 2021b)	TCYB23	53.3	66.5
11. ViPT (Zhu et al., 2023a)	CVPR23	66.3	74.5
12. CMDTrack (Zhang et al., 2025)	TPAMI25	65.7	74.8
13. LMTrack (Kang et al., 2025)	AAAI25	58.4	71.1
14. Mamba-FETrack (Huang et al., 2024)	PRCV24	59.4	71.8
15. Ours	–	62.6	76.9

Table 4
Component analysis on the FELT dataset. PCMF and Dyn. Temp. denote prompt-guided cross-modal fusion and dynamic template, respectively.

No.	Base	Prompt Generator	PCMF	Dyn. Temp.	SR	PR
1	✓				50.8	64.7
2	✓	✓			52.4	66.5
3	✓		✓		52.2	66.1
4	✓	✓	✓		53.3	67.6
5	✓	✓	✓	✓	**53.8**	**68.2**

Table 5
Ablation studies on the FELT dataset.

# Input Data	SR	PR	NPR
1. Event Frames only	46.5	59.6	56.6
2. RGB Frames only	51.8	65.6	63.1
3. RGB Frames & Event Frames	**53.8**	**68.2**	**65.2**
4. RGB Frames & Event Voxel	52.7	67.6	64.4
# Backbone Network	SR	PR	NPR
1. ViT-B	52.3	65.9	63.3
2. ViT-S	48.1	61.2	59.6
3. Vim-S	**53.8**	**68.2**	**65.2**
# Fusion Method	SR	PR	NPR
1. Addition	51.2	65.1	62.4
2. Concatenation	50.8	64.7	61.8
3. Prompt-guided Cross Fusion	**53.8**	**68.2**	**65.2**
# Fusion Position	SR	PR	NPR
1. y	52.6	66.9	63.9
2. Δ	52.3	66.6	63.8
3. B	52.1	66.1	63.4
4. C	**53.8**	**68.2**	**65.2**
# Prompt Generation	SR	PR	NPR
1. Random Initialization	52.2	66.1	63.2
2. w/o Prompt Pool	52.4	66.6	63.8
3. Prompt Generator	**53.8**	**68.2**	**65.2**

efficient long-range modeling, it may reduce bounding box accuracy under scale variation, leading to lower SR despite strong PR performance.

4.4. Ablation study

To comprehensively validate the effectiveness of each component in our proposed method, we conducted comprehensive ablation studies on the FELT dataset. Our analysis focuses on six key aspects: component analysis, input data, backbone architecture, fusion strategy, fusion position, and the Prompt Generator. The results of each ablation experiment are presented and discussed below.

• **Component Analysis.** As shown in Table 4, we progressively enhance the baseline model by incorporating the Prompt Generator and the Cross-modal Fusion strategy to assess their individual and combined contributions to performance. The baseline, which simply fuses RGB and event features via concatenation, achieves an SR of 50.8 and a PR of 64.7. Introducing the Prompt Generator alone improves the SR to 52.4 and the PR to 66.5. Similarly, integrating the prompt-guided Cross-modal Fusion (PCMF) module yields an SR of 52.2 and a PR of 66.1. When both modules are applied together, the model achieves an SR of 53.3 and a PR of 67.6. These results suggest that the Prompt Generator and PCMF are complementary: the former generates adaptive modality-specific prompts, while the latter enables these prompts to guide the counterpart modality inside the state-space branch. Therefore, the improvement is not merely from using two modalities, but from more effective RGB-event interaction enabled by the proposed prompt-guided cross-modal fusion design. Finally, we incorporate a template updating strategy that leverages dynamic templates to better capture the target's evolving appearance, resulting in further performance improvements.

• **Analysis on Input Data.** In the input modality analysis (top part of Table 5), we evaluate the tracking performance using only event frames, only RGB frames, and a fusion of both modalities. When using event frames alone, the performance is relatively low, with an SR of 46.5, PR of 59.6, and NPR of 56.6. In contrast, using only RGB frames yields better results. When both RGB and event frames are fused, the performance is further enhanced, reaching an SR of 53.8, PR of 68.2, and NPR of 65.2.

Table 6
Ablation study of key hyperparameters on the FELT dataset. Results are reported as SR/PR/NPR.

Hyperparameter	Setting 1	Setting 2	Setting 3	Setting 4
T	4 52.4/66.8/63.7	8 53.0/67.3/64.2	16 **53.8/68.2/65.2**	32 53.3/67.4/64.6
τ	0.1 53.4/67.8/64.8	0.3 **53.8/68.2/65.2**	0.5 53.2/67.3/64.3	0.7 53.2/67.1/64.4
k	4 53.1/67.2/64.2	8 **53.8/68.2/65.2**	10 53.5/68.0/65.2	12 53.3/67.6/64.8
r	1.2 53.6/67.7/64.9	1.5 **53.8/68.2/65.2**	1.8 53.2/67.5/64.4	2.0 52.9/67.0/63.8

These results suggest that the event modality provides complementary motion and high-dynamic-range cues, especially under challenging conditions such as fast motion and low-light scenes, thereby enhancing the robustness of RGB-Event tracking.

In addition, we further explored the asynchronous nature of event data by converting event streams into event voxel representations, which preserve richer temporal information. The experimental results show that the RGB Frames & Event Voxel setting outperforms the RGB-only baseline, but is slightly inferior to the combination of RGB frames and event frames. This indicates that fully exploiting such fine-grained temporal cues may require dedicated temporal modeling mechanisms.

• **Analysis on Backbone Network.** We further compare the performance of three backbone networks: ViT-B, ViT-S, and Vim-S. As shown in Table 5, ViT-B demonstrates solid performance, with an SR of 52.3, PR of 65.9, and NPR of 63.3. However, this comes at the cost of significantly increased parameter size and computational overhead. In contrast, Vim-S achieves better performance (SR: 53.8, PR: 68.2, NPR: 65.2) while maintaining substantially lower complexity. Therefore, to better balance accuracy and complexity, we ultimately selected Vim-S as the backbone network in our method.

• **Analysis on Fusion Method.** We evaluate three fusion strategies: Addition, Concatenation, and Prompt-guided Cross Fusion. Among them, our proposed Prompt-guided Cross Fusion achieves the best performance with SR of 53.8, PR of 68.2, and NPR of 65.2. This demonstrates that incorporating learnable, modality-specific prompts for cross-modal interaction provides more effective guidance for fusion than simple element-wise operations. These results demonstrate that the proposed prompt-guided fusion strategy is more effective than simple feature-level fusion in capturing cross-modal relationships between RGB and event modalities.

• **Analysis on Fusion Position.** To determine the optimal fusion strategy, we further explore different fusion positions in SSM matrices: y, Δ, B, and C. We do not inject prompts into matrix A because A is independent of the token sequence length, making it not naturally compatible with dynamic token-wise prompts. In addition, A governs the core state-transition dynamics of the SSM, modifying it may disturb the state evolution process and make optimization less stable. Among the explored fusion positions, fusion at position C matrix achieves the best performance (SR: 53.8, PR: 68.2, NPR: 65.2), suggesting that performing fusion on the output matrix provides the greatest benefit. We believe this operation enables each modality to effectively integrate and refine high-level features received from the other modality, resulting in more effective cross-modal interaction and ultimately enhancing tracking accuracy.

• **Analysis on Prompt Generator.** We then compared three different prompt generation strategies: Random Initialization, the absence of a Prompt Pool, and our proposed learnable Prompt Generator. As shown in Table 5, the learnable Prompt Generator consistently outperforms the other methods, demonstrating its effectiveness. This underscores the importance of modality-specific prompt learning in enhancing multimodal fusion by providing more adaptive and informative cross-modal guidance.

Table 7
Comparison of SR, parameters, FLOPs, and tracking speed on the FELT dataset.

No.	Tracker	Backbone	SR	Params (M)	FLOPs (G)	FPS
1	CEUTrack-B	ViT-B	52.3	97	1850	28
2	CEUTrack-S	ViT-S	48.1	60	1076	43
3	LMTrack	ViT-B	50.9	92	69	47
4	Mamba-FETrack	Vim-S	52.2	36	158	24
5	Ours	Vim-S	53.8	30	126	29

Table 8
Module-level latency analysis of Mamba-FETrack V2.

Component	Latency (ms)	Percentage
Projection Layer	1.1	3.3%
Prompt Generator	1.4	4.2%
FEMamba backbone	27.7	83.2%
FEMamba backbone (w/o PCMF)	27.3	–
Tracking head	3.1	9.3%
Total	**33.3**	**100.0%**

• **Hyperparameter Analysis.** As shown in Table 6, we investigate four key hyperparameters on the FELT dataset. Here, T denotes the number of basis prompts in the prompt pool, τ denotes the maximum-score threshold for low-confidence detection, k denotes the number of consecutive low-confidence frames, and r denotes the search region expansion ratio. The best overall performance is achieved with $T = 16$, $\tau = 0.3$, $k = 8$, and $r = 1.5$. A moderate prompt pool size provides sufficient modality-specific guidance while avoiding excessive redundancy. The threshold $\tau = 0.3$ effectively identifies unreliable tracking states, while $k = 8$ prevents overly frequent search-region expansion caused by short-term score fluctuations. Moreover, $r = 1.5$ provides an appropriately enlarged search area for target recovery without introducing excessive background distractors. Therefore, we adopt $T = 16$, $\tau = 0.3$, $k = 8$, and $r = 1.5$ as the default hyperparameter configuration in our method.

4.5. Efficiency analysis

To provide a more rigorous evaluation of Mamba-FETrack V2, Table 7 compares tracking performance, model size, FLOPs, and inference speed on the FELT dataset. Built upon the lightweight Vim-S backbone, our method achieves the highest SR of 53.8 with only 30M parameters, outperforming CEUTrack-B (52.3), LMTrack (50.9), and Mamba-FETrack (52.2), while maintaining the smallest model size among the compared methods. Compared with CEUTrack-S, which adopts a ViT-S backbone, our method achieves a substantial performance improvement with approximately half the number of parameters. Overall, Mamba-FETrack V2 achieves leading tracking performance on FE108 and FELT and the highest precision on COESOT, while requiring only 30M parameters and 126G FLOPs, demonstrating a favorable balance between tracking accuracy and computational complexity.

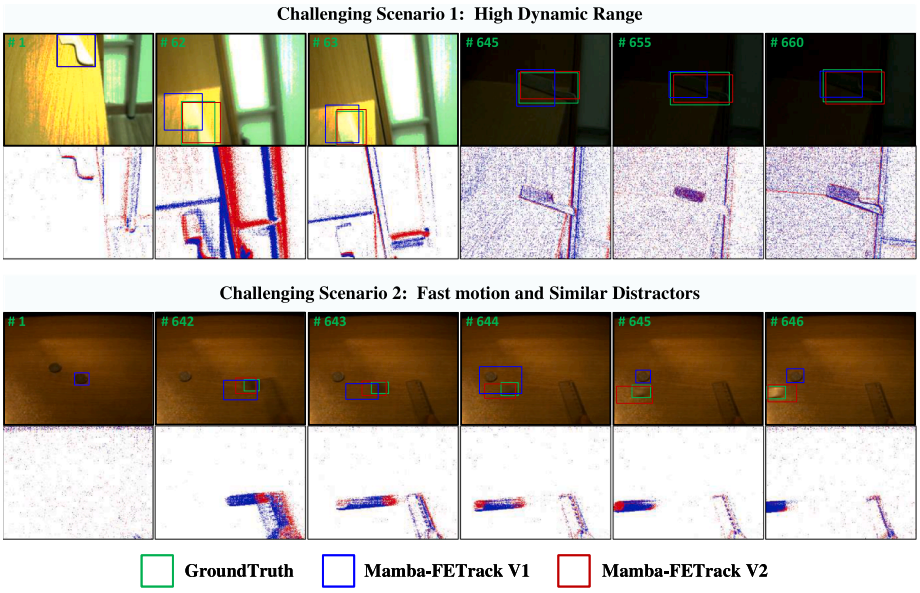


Fig. 4. Qualitative comparison between Mamba-FETrack V1 and Mamba-FETrack V2 under representative challenging scenarios.

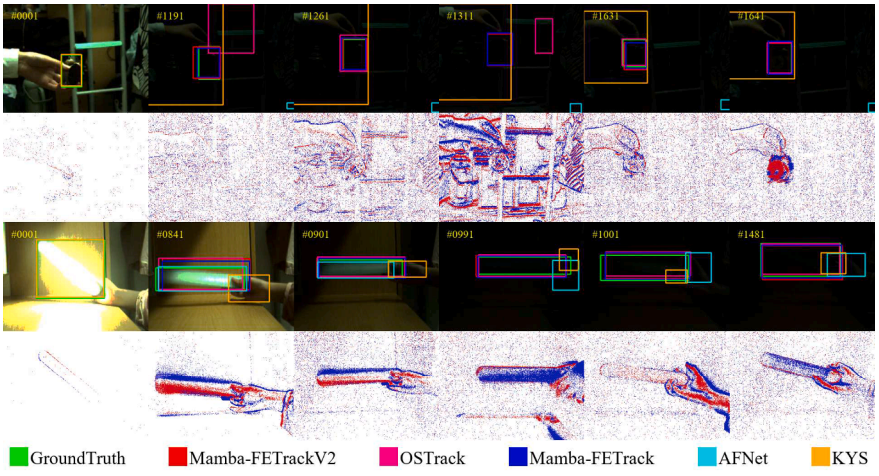


Fig. 5. Tracking results of our Mamba-FETrack V2 and other SOTA trackers.

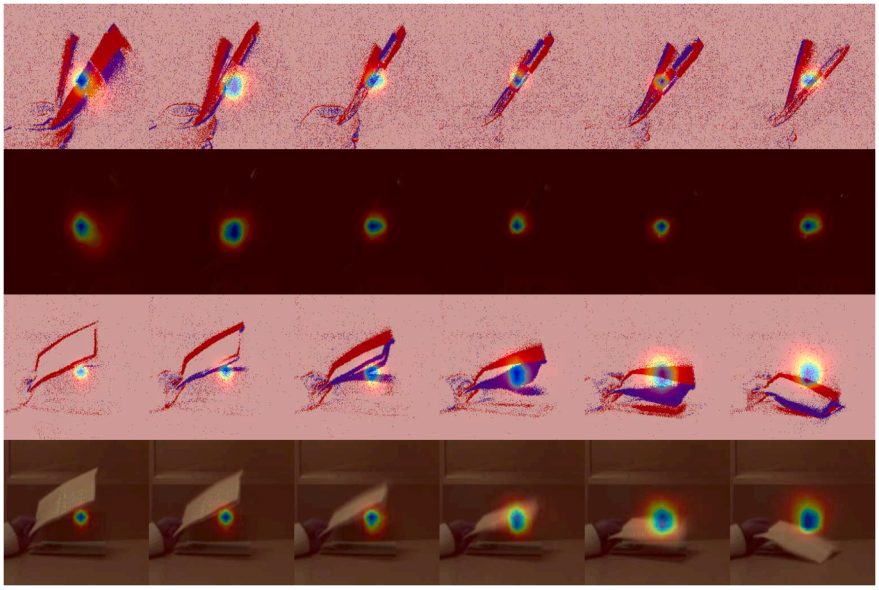


Fig. 6. Visualization of response maps generated by Mamba-FETrack V2.

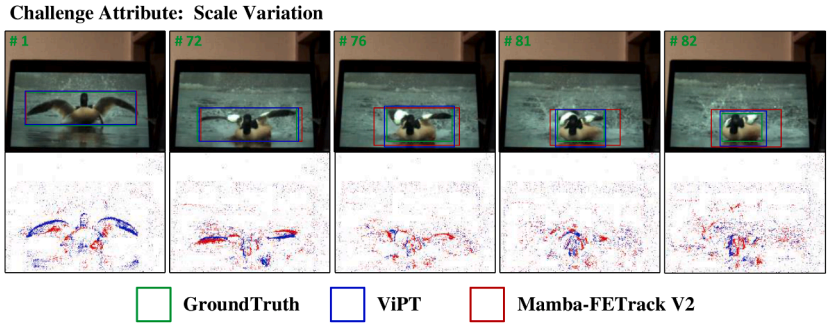


Fig. 7. Limitations of bounding box alignment under scale variation.

It should be noted that a smaller model size and lower theoretical computational complexity do not necessarily translate into faster inference on existing hardware. Although our method has the smallest model size among the compared trackers, its actual inference speed is lower than that of some Transformer-based methods. As further analyzed in Table 8, the FEMamba backbone accounts for 83.2% of the total network latency, suggesting that the primary runtime bottleneck is associated with its selective scanning operations. Compared with highly optimized matrix-multiplication and attention kernels, selective scanning is currently less amenable to efficient GPU parallelization and has not yet benefited from equally mature hardware-level optimization. Nevertheless, the proposed Prompt Generator and PCMF mechanism introduce only limited additional latency, demonstrating that the proposed cross-modal fusion design improves tracking performance with modest runtime overhead.

4.6. Qualitative comparison with the conference version

To further demonstrate the advantages of Mamba-FETrack V2 over our PRCV 2024 conference version, we provide qualitative comparisons under representative challenging scenarios in Fig. 4. The first sequence presents a high-dynamic-range condition, in which illumination changes degrade the RGB observations and lead to inaccurate predictions by Mamba-FETrack V1. The second sequence involves fast motion and similar distractors, under which V1 gradually drifts toward the distractor. By contrast, Mamba-FETrack V2 maintains more stable and accurate target localization in both scenarios. These results indicate that the newly introduced Prompt Generator and PCMF mechanism facilitate more effective cross-modal interaction, allowing the tracker to better exploit complementary RGB and event cues under challenging conditions.

4.7. Visualization

In addition to the comprehensive quantitative analysis provided, we also present qualitative visual results to offer a more intuitive understanding of the capabilities of Mamba-FETrack V2. Fig. 5 showcases a comparative visualization of our method alongside other trackers. It is evident that the bounding boxes predicted by Mamba-FETrack V2 maintain a closer and more consistent correspondence with the ground truth when contrasted with alternative approaches, particularly in scenarios involving complex motion or appearance changes. Furthermore, to illustrate the efficacy of our model in accurately focusing on the target, Fig. 6 displays the response maps predicted by Mamba-FETrack V2. These response maps reveal that for different object categories and various common objects depicted in the sequences, our method can effectively concentrate its learned features on the designated targets, distinguishing them from the surrounding environment. These visualization experiments serve to intuitively illustrate the superior tracking performance and precise target localization achieved by our proposed Mamba-FETrack V2 framework.

4.8. Limitation analysis

Despite the superior performance achieved by our method, there is still room for further improvement. On one hand, Mamba-FETrack V2 still has limitations in bounding box refinement under significant scale variation. As shown in Fig. 7, the model can localize the target center correctly but may not promptly adapt the predicted box to rapid scale changes. Unlike Transformer self-attention, which explicitly models dense token-to-token relationships, Mamba captures spatial interactions implicitly through sequential scanning. Therefore, subtle boundary and scale cues may be insufficiently modeled, leading to less precise bounding box alignment under scale variation. In future work, we will explore scale-aware refinement mechanisms, including multi-scale feature aggregation and adaptive bounding box regression, to improve the accuracy of target-scale estimation under significant scale variation. On the other hand, although the model has low computational complexity, its actual inference speed on GPUs remains limited. This is mainly because the selective scanning mechanism in Mamba is less amenable to GPU optimization and efficient parallelization. In future work, we will explore more parallel-friendly implementations and improved scanning strategies to further accelerate inference.

5. Conclusion and future work

In this paper, we present Mamba-FETrack V2, a novel RGB-Event-based visual object tracking framework that introduces a new paradigm for multimodal tracking. Specifically, we first design a Prompt Generator that dynamically produces two learnable prompt vectors by leveraging both modality-specific features and a shared prompt pool. Subsequently, the generated modality-specific prompt vectors and image embedding features are jointly processed by the proposed FEMamba module, which is built upon Vision Mamba, to facilitate prompt-guided cross-modal interaction and fusion. As a result, our algorithm achieves an optimal balance between tracking accuracy and computational complexity, establishing an efficient and high-performance solution for RGB-Event object tracking. In our future work, we plan to investigate more lightweight visual Mamba architectures to further enhance tracking efficiency. We believe this study will contribute to advancing Mamba-based object tracking research and inspire new developments in the field.

CRediT authorship contribution statement

Shiao Wang: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Conceptualization; **Ju Huang:** Writing – original draft, Visualization, Validation; **Qingchuan Ma:** Writing – original draft, Visualization, Validation; **Jinfeng Gao:** Writing – original draft, Visualization; **Chunyi Xu:** Writing – original draft, Visualization; **Xiao Wang:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Methodology; **Lan Chen:** Writing – review & editing, Writing – original draft, Visualiza-

tion; **Bo Jiang:** Writing – review & editing, Writing – original draft, Visualization, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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